S6 Proof Repair and Disproof Workbench — Private

Working mathematical notebook following the three-stream proof audit

26 August 2026

Abstract

This is a separate mathematical workbench for validating, repairing, or disproving the reconstructed compact complex threefold on S^6. The finite integer calculations are recorded as exact certificates: the monodromy invariant and coinvariant lattices, the finite-fibre indices, the parameter $p = 12\ell_0 - 4\ell_1 - 3\ell_2$ and Smith form $\mathrm{diag}(1, |p|)$, and the degree-two matrix with primitive left kernel $\Phi = (4, 2, 3, 2)$. The global conclusions remain **[CONDITIONAL]** until the printed matrices are identified with the actual geometric specialization, attaching, Mayer–Vietoris, and Leray maps.

The notebook names those bridge lemmas (C1–C3, V1, MV1, and L1), gives a conditional recognition theorem, and treats the Section 10/CDP issue in three separate questions: is a printed proof step valid, is the published theorem false, and does this candidate satisfy its hypotheses? It does not silently alter the reconstructed manuscript; the frozen 121-page audit and 108-page transcription remain under `audit_baseline/`.

For the Section 10 route, the source proof of the candidate's compact smooth analytic geometry and full cusp fibre has now been independently retraced. The conductor-supported differential, compact-Cartier-fibre duality, and top-degree base change then prove $R^2 f_*(T_X \otimes L) \neq 0$ for every $L \in \mathrm{Pic}^0(X)$, without assuming the unresolved equality $H^2(X; \mathbb{Z}) = 0$. Extension to every component of $\mathrm{Pic}(X)$ is equivalent for this method to the separate open restriction-map statement $\eta^* i^* = 0$. None of this promotes the global topology or by itself declares the published CDP theorem false.

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1 Problem, evidence boundary, and status language

The target is the manuscript's claimed compact complex threefold X equipped with a holomorphic map $f\colon X \to \mathbb{P}^1$, with

$$X \cong_{\mathrm{diff}} S^6, \qquad a(X) = 1, \qquad \langle c_3(T_X), [X] \rangle = 2.$$

The review found no finite-algebra contradiction in the completed manuscript, but it did not certify the geometric inputs from which the terminal matrices are obtained.

1.1 Status rules

[VERIFIED] Reproduced exactly or proved in this notebook from stated hypotheses.

[CONDITIONAL] Correct after explicitly named geometric or analytic hypotheses are granted.

[OPEN] A load-bearing statement for which no complete proof or counterexample is yet recorded here.

[REFUTED] A statement defeated by a printed counterexample or a complete contradiction.

1.2 Current ledger

Status	Object	Present evidence
[VERIFIED]	terminal finite algebra	Monodromy products/orders, Smith forms, exterior-power ranks, displayed lattice saturation, period identities, $\eta^2 = 12$, and the Gram determinant have exact checks in the retained review harnesses.
[VERIFIED]	candidate analytic geometry	Sections 3–6 construct the smooth compact connected threefold, proper holomorphic map, full reduced normal-crossings cusp fibre, dP_6 normalization, and a trivial canonical bundle on its open cusp neighbourhood. The source proof chain has been independently retraced.

Status	Object	Present evidence
[VERIFIED]	conductor differential	The normalization–conductor sequence and the local x, xy, and xyz models produce the nonzero ambient differential under the stated reduced normal-crossings and pullback-triviality hypotheses.
[VERIFIED]	compact-Cartier duality	Ramis–Ruget duality, its Gorenstein complete-intersection calculation, Cartier adjunction, and compact coherent-cohomology finiteness give the stated perfect pairing without a normality hypothesis.
[VERIFIED]	top-degree base change	The analytic cohomology-and-base-change theorem gives β_c^2 as an isomorphism when H^3 of the fibre vanishes; compactness, smoothness, connectedness, and nonconstancy make properness and flatness automatic over a smooth curve.
[VERIFIED]	Section 10 on Pic^0	Local canonical triviality and $c_1(L) = 0$ show $\eta^* A_L \simeq \mathcal{O}_{\mathrm{dP}_6}$ for every $L \in \mathrm{Pic}^0(X)$. Conductor descent, duality, and base change give $R^2 f_*(T_X \otimes L) \neq 0$ for every such L.
[CONDITIONAL]	$\pi_1(X)$ and $H_*(X;\mathbb{Z})$	Correct for the printed presentations and boundary matrices; their equality with the maps induced by the analytic gluings is not yet established here.
[OPEN]	cusp collapse	The analytic action, quotient, and central-fibre charts are verified. C1 still requires a finite equivariant CW model and a global strong deformation retraction compatible with all strata.
[OPEN]	integral topology maps	Specialization, transfer indices, saturation, torsion, primitive bases, orientations, and degree-three signs.
[OPEN]	all-line-bundle extension	For this conductor method, extension from $\mathrm{Pic}^0(X)$ to all of $\mathrm{Pic}(X)$ is equivalent to $r = \eta^* i^* = 0$. The claimed $H^2(X;\mathbb{Z}) = 0$ is sufficient but remains conditional on the topology bridges.
[OPEN]	CDP adjudication	The printed cotangent-sheaf reduction misses the conductor-supported section. A line-by-line repair search and the candidate's remaining global CDP hypotheses must be settled before declaring Theorem 2.2 false.

The working rule is asymmetric: a failed proof step does not refute the theorem, and a coherent terminal calculation does not prove that the required geometric map exists.

2 Fixed algebraic data reproduced by the audits

This section freezes the exact statements that need no further numerical experimentation. The research task is to prove that the geometric maps have these algebraic representatives.

Let

$$H_1(F;\mathbb{Z}) = \Lambda = \langle \widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta} \rangle, \qquad H^1(F;\mathbb{Z}) = V = \langle \gamma, u, w, \delta \rangle.$$

Proposition 2.1 (checked monodromy lattices). *[VERIFIED] The displayed monodromy matrices satisfy*

$$A_1 A_2 M_0 = I, \qquad \ker(M_0 - I) = \mathrm{im}(M_0 - I) = \Lambda_{\mathrm{tor}} := \langle \widehat{w}, \widehat{\delta} \rangle.$$

For $T_0 = I + N$ with $N^2 = 0$, the invariant lattices in degrees one through three are

$$\begin{aligned} V^{T_0} &= \langle \gamma, u \rangle, \\ (\wedge^2 V)^{T_0} &= \langle \gamma u, \gamma\delta, uw, \gamma w - u\delta \rangle, \\ (\wedge^3 V)^{T_0} &= \langle \gamma uw, \gamma u\delta \rangle. \end{aligned}$$

They are saturated and have ranks $(1, 2, 4, 2, 1)$ across all degrees. For $G = \langle T_1, T_2 \rangle$, the nonzero coinvariant lattices used downstream are

$$V_G = \mathbb{Z}\bar{\delta}, \qquad (\wedge^2 V)_G = \mathbb{Z}[\gamma\delta], \qquad (\wedge^3 V)_G = \mathbb{Z}[uw\delta],$$

all torsion-free, with $[uw] = 6[\gamma\delta]$.

Proposition 2.2 (checked finite-fibre indices)**.** ***[VERIFIED]*** *In degrees $q = 0, 1, 2, 3, 4$, the two finite-fibre specialization-index tuples in the manuscript's bases are*

$$(1, 3, 1, 1, 3), \qquad (1, 4, 2, 2, 4).$$

For $v_2 = -\varepsilon'$, the degree-three index-two quotient at p_2 is generated by $[\gamma uw]$.

Proposition 2.3 (checked group presentation)**.** ***[VERIFIED]*** *Once the asserted based inclusions are granted, the presentation reduces to*

$$\left\langle c, x, y \;\middle|\; c \text{ central},\ xy = c^{\ell_0},\ x^3 = c^{\ell_1},\ y^4 = c^{\ell_2} \right\rangle.$$

After eliminating y, its relation matrix is

$$R = \begin{pmatrix} 3 & -\ell_1 \\ 4 & \ell_2 - 4\ell_0 \end{pmatrix}, \qquad \det R = -p, \qquad p = 12\ell_0 - 4\ell_1 - 3\ell_2,$$

and its Smith form is $\operatorname{diag}(1, |p|)$.

Proposition 2.4 (checked degree-two terminal matrix)**.** ***[VERIFIED]*** *The displayed degree-two matrix has columns*

$$(2, -4, -2, 3),\ (1, -2, -2, 3),\ (3, 0, -4, 0),\ (0, 1, 0, -1),\ 0,\ 0.$$

Its image has rank three and is saturated, its cokernel is $\mathbb{Z}$, and its primitive left-kernel generator is

$$\Phi = (4, 2, 3, 2).$$

Proposition 2.5 (checked all-degree terminal homology package)**.** ***[VERIFIED]*** *For $p \neq 0$, the retained exact audit applied to the manuscript's printed degree-by-degree Mayer–Vietoris/Leray data gives the terminal abelian-group tuple*

$$(H_0, H_1, H_2, H_3, H_4, H_5, H_6) \cong (\mathbb{Z}, \mathbb{Z}/|p|\mathbb{Z}, \mathbb{Z}/|p|\mathbb{Z}, \mathbb{Z}/|p|\mathbb{Z}, \mathbb{Z}/|p|\mathbb{Z}, 0, \mathbb{Z}).$$

Consequently the printed algebraic package has the homology of S^6 when $p = -1$. The exact evidence is frozen in `audit_baseline/annotated_audit.tex`, ledger item B-EX-06, and in the degree-by-degree calculation surrounding Theorem 7.22 of `audit_baseline/clean_transcription.tex`.

Every proposition in this section concerns the printed matrices and lattices. None identifies those matrices with a geometric specialization, attaching, Mayer–Vietoris, or Leray map.

3 The prior Fable certificate: relevance and boundary

The separate Erdős–Strauss workspace contains a pinned copy of Jon Poplett's JGPTech package for the polynomial map publicly attributed there to Fable. This is relevant as a standard of verification, not as evidence for any step of the present S^6 construction. The canonical checkout and its immutable mirror are listed with exact absolute paths in `comparative_context/README.md`; within either checkout the relevant file is `JGPTech_Fun/Jacobian/aceg.py`.

Define $F = (F_1, F_2, F_3)\colon \mathbb{C}^3 \to \mathbb{C}^3$ by

$$\begin{aligned} F_1(x,y,t) &= (1+xy)^3 t + y^2(1+xy)(4+3xy), \\ F_2(x,y,t) &= y + 3x(1+xy)^2 t + 3xy^2(4+3xy), \\ F_3(x,y,t) &= 2x - 3x^2 y - x^3 t. \end{aligned}$$

Proposition 3.1 (replayed finite certificate)**.** ***[VERIFIED]*** *Exact rational-polynomial arithmetic gives*

$$\det DF = -2.$$

Moreover, the three distinct points

$$\left(0,0,-\tfrac{1}{4}\right), \qquad \left(1,-\tfrac{3}{2},\tfrac{13}{2}\right), \qquad \left(-1,\tfrac{3}{2},\tfrac{13}{2}\right)$$

all map to $(-\frac{1}{4},0,0)$*. The local pinned checker `Jacobian/aceg.py` was replayed successfully on 26 August 2026 with the 'base' and manifest checks. Its exact SHA-256, mirror commit, and absolute paths are recorded in `comparative_context/README.md`. Thus the displayed map itself has the advertised constant nonzero Jacobian and noninjective fibre.*

Remark 3.2 (provenance boundary). The located corpus is an independently organized proof and exact audit of the displayed map, not an original model-session transcript. No original Fable/Claude model-session source was located. None of this certificate supplies a lemma about the complex-threefold manuscript, and the Jacobian counterexample is kept distinct from Erdős–Strauss arithmetic.

The transferable lesson is procedural. The prior result terminates in a small certificate—one determinant identity and one explicit collision—that can be checked independently of its discovery narrative. The present paper will reach the same evidentiary level only when its analytic gluings terminate in independently reconstructed integral chain maps. Accordingly this workbench uses typed claims, logical dependency edges, typed morphisms, explicit uncertainties, clean-archive replays, and rendered-page QA; see `comparative_context/README.md`.

4 Topology: exact reduction and geometric obligations

4.1 What the exact arithmetic already proves

Let

$$p = 12\ell_0 - 4\ell_1 - 3\ell_2.$$

The retained exact harness verifies the determinant and Smith-normal-form calculations for the printed relation matrix. In particular, the manuscript's chosen parameters $(\ell_0, \ell_1, \ell_2) = (0, 1, -1)$ give $p = -1$.

Proposition 4.1 (terminal group calculation)**.** ***[VERIFIED]*** *For the printed based presentation and its stated meridian relations, the resulting group is cyclic of order* $|p|$*. Hence the printed presentation is trivial for* $(0, 1, -1)$*.*

Proof. The exact row and column operations recorded in the audit reduce the displayed 2×2 relation matrix to Smith normal form $\operatorname{diag}(1, |p|)$. A group presented by that matrix is therefore $\mathbb{Z}/|p|\mathbb{Z}$, which is trivial when $p = -1$. □

Remark 4.2. Proposition 4.1 is a theorem about the printed presentation. It becomes a theorem about $\pi_1(X)$ only after the based inclusions induced by all three analytic attaching maps have been derived in the same generators and orientations.

4.2 The map that must be identified

Write the relevant decomposition schematically as $X = U \cup N$, where N is the union of the filling pieces and $U \cap N$ is the disjoint union of their collars. The homological boundary map is

$$\delta_q = (i_{U*}, -i_{N*}) \colon H_q(U \cap N; \mathbb{Z}) \longrightarrow H_q(U; \mathbb{Z}) \oplus H_q(N; \mathbb{Z}).$$

The printed matrices denoted α_q have verified ranks, Smith forms, and saturation properties. The central question is not another Smith-form computation; it is the equality

$$\boxed{\alpha_q = [\delta_q]_{\mathcal{B}_q}}$$

for explicitly constructed primitive oriented integral bases $\mathcal{B}_q$.

The equality is decomposed below into the six nonredundant proof obligations C1, C2, C3, V1, MV1, and L1. The first three construct the finite equivariant model, compute its integral collapse map, and identify that map with nearby-cycle specialization. Obligation V1 fixes the based van Kampen inclusions; MV1 identifies the actual Mayer–Vietoris boundary cylinders and multiple-fibre indices; L1 supplies an independently oriented Leray–Mayer–Vietoris comparison. This crosswalk is exhaustive for the conditional endpoint stated here; none of the six is replaced by matching a rank or by choosing the desired matrix as a normalization.

Disproof test 4.3 (fast falsification targets)**.** Any of the following disproves the claimed topology calculation: a nonprimitive geometric image where α_q assumes a primitive one; an index or torsion class omitted by specialization; a meridian sign changing $p = -1$ to a nonunit; or a Leray transgression differing from $\pm p\omega$ by a nonunit factor.

4.3 Conditional recognition

Theorem 4.4 (precise conditional endpoint)**.** ***[CONDITIONAL]*** *Assume obligations C1–C3, V1, MV1, and L1 are discharged and give exactly the printed integral maps. Then the verified terminal calculations yield*

$$\pi_1(X) = 0, \qquad H_k(X; \mathbb{Z}) = \begin{cases} \mathbb{Z}, & k = 0, 6, \\ 0, & 1 \le k \le 5. \end{cases}$$

Consequently X is a smooth homotopy 6-sphere; the standard fact $\Theta_6 = 0$ identifies it diffeomorphically with S^6.

Proof. The hypotheses identify the actual van Kampen and Mayer–Vietoris/Leray maps with the already verified presentations and matrices. Their Smith forms give the displayed fundamental group and homology. A simply connected integral homology sphere is a homotopy sphere by Hurewicz and Whitehead; the final smooth recognition uses $\Theta_6 = 0$. □

5 First proof attacks: exact lemmas to establish or break

The following statements turn the review's broad objections into falsifiable mathematical targets. A proof must derive the maps from the analytic charts; matching ranks is insufficient.

5.1 Cusp quotient and collapse

The source defines an infinite toric threefold Y from the height-one cones over the periodic A_2 triangulation and an action of the form

$$\Psi_{\bar\lambda}(x,t) = \big(c_{\bar\lambda}(t)t^{B_0\bar\lambda}x, t\big), \qquad \bar\lambda \in \bar\Lambda.$$

Theorem 4.5's analytic part has now been independently retraced: this action is free and properly discontinuous, its quotient is a smooth Hausdorff threefold proper over the disc, and its central fibre has the asserted local normal-crossings equations. C1 below is narrower: it asks for the finite equivariant topological model and collapse that are not supplied by those analytic quotient facts.

Proof obligation 5.1 (C1: finite equivariant cusp model)**.** Construct a locally finite $\bar\Lambda$-equivariant CW model of $(Y_{\le\eta}, Y_0)$ whose quotient is finite, and prove that the collapse is equivariant in the diagram

$$\begin{array}{ccc} Y_{\le\eta} & \xrightarrow{r_Y} & Y_0 \\ \big\downarrow{\scriptstyle /\bar\Lambda} & & \big\downarrow{\scriptstyle /\bar\Lambda} \\ N_0' & \xrightarrow{r} & W. \end{array}$$

The proof may cite the verified analytic action theorem, but it must still establish continuity of the proposed collapse across the one-, two-, and three-branch strata, compatibility with the fibre-torus action, finiteness of the quotient CW model, and a strong deformation retraction.

Proof obligation 5.2 (C2: integral collapse matrix)**.** With fixed cellular orientations, prove for $0 \le q \le 4$ the exact sequence

$$0 \longrightarrow \operatorname{im}(\wedge^q M_0 - I) \longrightarrow \wedge^q\Lambda \xrightarrow{c_*} H_q(W;\mathbb{Z}) \longrightarrow 0.$$

In particular, derive $c_*\colon \Lambda \to H_1(W) = \bar\Lambda$ as the quotient map, prove

$$c_*(\widehat\gamma \wedge \widehat u) = [\mathbb{T}_B], \qquad c_*(\widehat w \wedge \widehat\delta) = 0,$$

and record the degree-three orientations rather than inferring them from ranks.

Proof obligation 5.3 (C3: collapse equals specialization)**.** Prove functorially and integrally that

$$c^* = i_F^* r^* = \mathrm{sp}^q_{\mathrm{nc}}, \qquad \operatorname{im} c^* = (\wedge^q V)^{T_0},$$

including saturation, torsion, bases, and the degree-three sign. The comparison diagram must state the precise nearby-cycle edge map; the equality may not be defined by choosing the manuscript's answer as a normalization.

5.2 Based inclusions

Choose one basepoint on the zero section and fixed whiskers through every collar.

Proof obligation 5.4 (V1: primitive meridians and drift)**.** Derive from the coordinate gluings, with no unrecorded conjugations,

$$\widehat g_j^{\,m_j} = t_{v_j}, \qquad \widetilde a_0 = t_\mu \widehat\rho_2^{-1}\widehat\rho_1^{-1}.$$

Prove that the clockwise meridian lifts to $\widehat g_j$ rather than its inverse or a nonprimitive power, that the two finite-fibre signs agree with the chosen orientations, that the zero-section whiskers add no hidden translation, and that the tautological cusp disc gives $\mu = 0$.

5.3 Mayer–Vietoris and Leray

Proof obligation 5.5 (MV1: geometric boundary cylinders)**.** Using the independently constructed collars and the maps from C2–C3, prove for every relevant degree q that

$$\alpha_q = \big[(i_{U*}, -i_{N*}) : H_q(U \cap N; \mathbb{Z}) \longrightarrow H_q(U; \mathbb{Z}) \oplus H_q(N; \mathbb{Z})\big]_{\mathcal{B}_q}$$

in the manuscript's named primitive oriented bases. This must include a chain-level check of every zero column, sign, torsion class, and saturation claim; equality of ranks or Smith forms is not sufficient. For $K_k = \ker(H_k(M) \to H_k(W))$, derive geometrically

$$K_1 = \mathbb{Z}[\widetilde{a}_0], \qquad K_2 = \langle \xi_{\widehat{w}}, \xi_{\widehat{\delta}} \rangle,$$

and fix one orientation convention for

$$\partial(j_* \xi_\nu) = (I - A_1^{-1})\nu = (I - A_2)\nu.$$

The same calculation must recover the multiple-fibre class $[F] \longmapsto (3[S_1], -4[S_2])$ and the index-two degree-three image at p_2.

Proof obligation 5.6 (L1: transgression comparison)**.** Construct the clutching cocycle that compares the Leray transgression with the Mayer–Vietoris obstruction. For 12γ, derive the discrepancy triple

$$(4\ell_1 - \alpha_1,\ 3\ell_2 - \alpha_2,\ -(12\ell_0 - \alpha_1 - \alpha_2)),$$

whose sum is $-p$, and thereby determine—rather than assume—the sign in

$$d_2^{0,1}(12\gamma) = \pm p\,\omega.$$

Then prove that the cup-product and coinvariant identifications are natural in the same orientations before using multiplicativity in higher degrees.

Until C1–C3, V1, MV1, and L1 are established, the conclusion $X \cong S^6$ remains conditional. The integer algebra downstream of these statements is already fixed by the previous section.

6 The non-normal fibre: exact claim under investigation

Status convention. This notebook does not assert that the CDP theorem is false. It studies one specific inference in the printed proof and the conditional application of a conductor-supported differential to the candidate fibre W. A defect in the printed inference, an alternative proof of the published theorem, and failure of the candidate construction are logically distinct possibilities.

6.1 The local conductor proposition

Proposition 6.1 (conductor-supported ambient differential)**.** ***[VERIFIED]*** *Let $g : \mathcal{X} \to \Delta$ be proper, with $\mathcal{X}$ a smooth complex threefold, and let $S = g^{-1}(0)$ be a reduced normal-crossings surface. Write $\eta : \widetilde{S} \to S$ for the normalization, $D \subset S$ for the reduced singular locus, and $\widetilde{D} = \eta^{-1}(D)$. Let A be a line bundle on S such that $\eta^* A \simeq \mathcal{O}_{\widetilde{S}}$, and choose a nowhere-vanishing section e of $\eta^* A$. Then there is a nonzero section*

$$s \in H^0\Big(S, \Omega^1_{\mathcal{X}}|_S \otimes A\Big)$$

whose image in $\Omega^1_S \otimes A$ is supported on D, hence is torsion, and whose image in

$$\widetilde{\Omega}^1_S \otimes A, \qquad \widetilde{\Omega}^1_S := \Omega^1_S / \mathcal{T}(\Omega^1_S),$$

is zero. If S is connected and $A \not\cong \mathcal{O}_S$, then $H^0(S, N^*_{S/\mathcal{X}} \otimes A) = H^0(S, A) = 0$, so s does not come from the conormal term.

Proof. Locally $g = ux$, uxy, or $uxyz$, where u is a unit. The conormal map is generated by $dg|_S$. At a double point,

$$dg|_S = u(y\,dx + x\,dy),$$

and at a triple point it has coefficients in (xy, xz, yz). Thus $dg|_S$ is nonzero away from D but restricts to zero on D; equivalently, it lies in the conductor ideal times $\Omega^1_{\mathcal{X}}|_S$.

The normalization–conductor square gives

$$0 \longrightarrow \mathcal{O}_S \longrightarrow \eta_*\mathcal{O}_{\widetilde{S}} \oplus \mathcal{O}_D \longrightarrow \eta_*\mathcal{O}_{\widetilde{D}} \longrightarrow 0.$$

After tensoring with the locally free sheaf $E = \Omega^1_{\mathcal{X}}|_S \otimes A$, the compatible pair

$$\big(\eta^*(dg|_S) \otimes e,\, 0\big)$$

descends uniquely to s. It is nonzero off D. Its intrinsic differential vanishes off D, so its image in $\Omega^1_S \otimes A$ is conductor-supported torsion. For nontrivial A, a global section of A pulls back to constants on the compact connected components of $\widetilde{S}$. Compatibility across the conductor and connectedness of S force either all constants to vanish or all to be nonzero; the latter would trivialize A. Hence $H^0(S, A) = 0$. □

Consequently the implication

$$\Big[H^0(S, N^*_{S/\mathcal{X}} \otimes A) = 0, \quad H^0(S, \widetilde{\Omega}^1_S \otimes A) = 0\Big] \Longrightarrow H^0(S, \Omega^1_{\mathcal{X}}|_S \otimes A) = 0$$

is not a formal consequence of the conormal sequence for a non-normal normal-crossings surface. The official CDP article prints precisely this reduction on p. 692 [2, p. 692]. This observation concerns that inference, not by itself the truth value of the theorem.

Lemma 6.2 (pullback triviality on a dP_6 normalization)**.** ***[VERIFIED]*** *Let $i : W \hookrightarrow X$ be a closed complex subspace with $H^2(X; \mathbb{Z}) = 0$, and suppose the normalization $\eta : \widetilde{W} \to W$ satisfies $\widetilde{W} \simeq \mathrm{dP}_6$. Then, for every $B \in \mathrm{Pic}(X)$,*

$$\eta^* i^* B \simeq \mathcal{O}_{\widetilde{W}}.$$

Proof. The group containing $c_1(B)$ is zero, so $c_1(\eta^* i^* B) = 0$. Since $H^1(\mathrm{dP}_6, \mathcal{O}) = 0$, the exponential sequence makes

$$c_1 : \mathrm{Pic}(\mathrm{dP}_6) \longrightarrow H^2(\mathrm{dP}_6; \mathbb{Z})$$

injective. Hence the pullback line bundle is trivial. □

6.2 Candidate geometry and the topology-free Pic^0 route

Proposition 6.3 (analytic geometry of the reconstructed cusp fibre)**.** ***[VERIFIED]*** *The reconstructed source produces a connected compact smooth complex threefold X, a proper surjective holomorphic map $f : X \to B \simeq \mathbb{P}^1$, and a full scheme-theoretic fibre $W = f^{-1}(p_0)$ with the following properties:*

(i) W is a connected reduced normal-crossings Cartier surface, with local equations z_0, $z_0 z_1$, and $z_0 z_1 z_2$ at its one-, two-, and three-branch points;

(ii) its normalization is $\eta : \mathrm{dP}_6 \to W$;

(iii) W has an open neighbourhood $N_0 \subset X$ on which ω_X is trivial.

Proof. The period construction in Theorem 3.4 of the frozen transcription supplies the admissible family and the cusp-holomorphic matrix $C(t)$. In source pp. 26–28 (lines 1331–1446), Theorem 4.5 proves that the toric action is free and properly discontinuous, that its quotient N_0 is a connected Hausdorff smooth threefold, that $f_0 : N_0 \to \Delta$ is proper, and that its full fibre is reduced with exactly the three displayed local equations. The same proof constructs an invariant nowhere-vanishing holomorphic three-form on N_0.

Proposition 4.6, source p. 29 (lines 1453–1465), proves that the finite map $\mathrm{dP}_6 \to W$ is surjective and generically injective and hence is the normalization. Proposition 4.7(i)–(iii) and Theorem 5.4 give the holomorphic overlap maps. Finally Theorem 6.2, source pp. 36–37 (lines 1811–1834), checks the quotient atlas, Hausdorffness, properness, compactness, and connectedness, and identifies $f^{-1}(D_0) = N_0$ and $f^{-1}(p_0) = W$ as a reduced divisor. Because the embedding $N_0 \hookrightarrow X$ is open and holomorphic, the constructed trivialization of K_{N_0} is a trivialization of $\omega_X|_{N_0}$.

These source gates were rechecked independently in the present workbench; no topology, Mayer–Vietoris, or Leray assertion is used here. □

Proposition 6.4 (Pic^0 pullback without global topology)**.** ***[VERIFIED]*** *Let X be a compact complex manifold, let $i : W \hookrightarrow X$ have normalization $\eta : \widetilde{W} \to W$ with $\widetilde{W} \simeq \mathrm{dP}_6$, and suppose ω_X is trivial on a neighbourhood of W. For every $L \in \mathrm{Pic}^0(X)$, put*

$$A_L = (L^\vee \otimes \omega_X)|_W.$$

Then

$$\eta^* A_L \simeq \mathcal{O}_{\widetilde{W}}.$$

Proof. Every $L \in \mathrm{Pic}^0(X)$ has $c_1(L) = 0$. Therefore $c_1(\eta^* i^* L) = 0$. Since $H^1(\mathrm{dP}_6, \mathcal{O}) = 0$, the exponential sequence makes the Chern-class map on $\mathrm{Pic}(\mathrm{dP}_6)$ injective, so $\eta^* i^* L$ is trivial. The neighbourhood trivialization of ω_X also makes $\eta^* i^* \omega_X$ trivial, proving the formula for A_L. □

For the stronger all-line-bundle assertion, define

$$r := \eta^* i^* : \mathrm{Pic}(X) \longrightarrow \mathrm{Pic}(\mathrm{dP}_6).$$

The preceding proof gives $r|_{\mathrm{Pic}^0(X)} = 0$ and the local canonical trivialization gives $r(\omega_X) = 0$. Hence, for arbitrary L,

$$\eta^* A_L \simeq r(L)^\vee.$$

Thus the present conductor method extends from $\mathrm{Pic}^0(X)$ to every $L \in \mathrm{Pic}(X)$ exactly when $r = 0$. The still conditional equality $H^2(X; \mathbb{Z}) = 0$ is one sufficient route to $r = 0$ by Lemma 6.2, but it is no longer needed for the Pic^0 statement relevant to the generic twists in CDP20.

Proposition 6.5 (candidate conductor application on Pic^0)**.** ***[VERIFIED]*** *For every $L \in \mathrm{Pic}^0(X)$, putting*

$$\mathcal{F}_L = T_X \otimes L, \qquad A_L = (L^\vee \otimes \omega_X)|_W,$$

one has

$$H^0(W, \Omega^1_X|_W \otimes A_L) \neq 0.$$

Proof. Proposition 6.3 supplies the proper smooth family, the full reduced normal-crossings fibre and its dP_6 normalization. Proposition 6.4 gives $\eta^* A_L \simeq \mathcal{O}_{\mathrm{dP}_6}$. The verified conductor proposition then supplies the nonzero section. □

Corollary 6.6 (conditional extension to all line bundles)**.** ***[CONDITIONAL]*** *If $r = 0$, the conclusion of Proposition 6.5 holds for every $L \in \mathrm{Pic}(X)$.*

6.3 The exact duality implication required

Proposition 6.7 (absolute duality on a compact Cartier fibre)**.** ***[VERIFIED]*** *Let $f : X \to B$ be a holomorphic map from a smooth complex threefold to a smooth complex curve. Suppose that $i : W = f^{-1}(b) \hookrightarrow X$ is compact, is the full scheme-theoretic fibre, and is an effective Cartier divisor. For every line bundle L on X, put*

$$\mathcal{F}_L = T_X \otimes L, \qquad A_L = (L^\vee \otimes \omega_X)|_W.$$

Then W is a pure two-dimensional Gorenstein complex space and there is a functorial perfect pairing

$$H^2(W, i^*\mathcal{F}_L) \times H^0(W, \Omega^1_X|_W \otimes A_L) \longrightarrow \mathbb{C}.$$

Equivalently,

$$H^2(W, i^*\mathcal{F}_L)^\vee \simeq H^0(W, \Omega^1_X|_W \otimes A_L).$$

No normality hypothesis on W is required.

Proof. Because W is an effective Cartier divisor in the smooth threefold X, each local ring of W is a two-dimensional hypersurface quotient of a regular local ring. Thus W is Cohen–Macaulay and Gorenstein. If $K_W^\bullet$ denotes the analytic dualizing complex, the Gorenstein criterion and complete-intersection computation of Ramis–Ruget give

$$K_W^\bullet \simeq \omega_W[2]$$

for an invertible dualizing sheaf ω_W [4, Proposition 2 and the complete-intersection calculation, pp. 90–91].

Cartier adjunction follows by applying $\mathcal{H}om_X(-, \omega_X)$ to

$$0 \longrightarrow \mathcal{O}_X(-W) \longrightarrow \mathcal{O}_X \longrightarrow i_*\mathcal{O}_W \longrightarrow 0 :$$

$$\omega_W = \mathcal{E}xt^1_X(i_*\mathcal{O}_W, \omega_X) \simeq (\omega_X \otimes \mathcal{O}_X(W))|_W.$$

Since $W = f^*(b)$ as a scheme-theoretic Cartier divisor, $\mathcal{O}_X(W) \simeq f^*\mathcal{O}_B(b)$. Its restriction to W is the pullback of the one-dimensional vector space $\mathcal{O}_B(b)|_b$, so a choice of nonzero vector gives

$$\mathcal{O}_X(W)|_W \simeq \mathcal{O}_W, \qquad \omega_W \simeq \omega_X|_W.$$

The choice changes the final pairing by a nonzero scalar only.

Ramis–Ruget duality pairs, for coherent $\mathcal{G}$,

$$H^p_c(W, \mathcal{G}) \quad \text{with} \quad \mathrm{Ext}^{-p}_W(\mathcal{G}, K_W^\bullet)$$

[4, Theorem 2, pp. 82–83]. Compactness gives $H^p_c(W, \mathcal{G}) = H^p(W, \mathcal{G})$. With $p = 2$ and $\mathcal{G} = i^*\mathcal{F}_L$, the shift above gives

$$\mathrm{Ext}^{-2}_W(\mathcal{G}, K_W^\bullet) \simeq \mathrm{Hom}_W(\mathcal{G}, \omega_W).$$

The coherent cohomology groups are finite-dimensional by Grauert finiteness [7]; in particular their natural topologies are separated, so the strong duality of Ramis–Ruget is an ordinary algebraic perfect duality. Since $\mathcal{G}$ is locally free,

$$\begin{aligned} \mathrm{Hom}_W(\mathcal{G}, \omega_W) &= H^0(W, \mathcal{G}^\vee \otimes \omega_W) \\ &\simeq H^0(W, \Omega^1_X|_W \otimes L^\vee|_W \otimes \omega_X|_W) \\ &= H^0(W, \Omega^1_X|_W \otimes A_L). \end{aligned}$$

The pairing is cup product followed by evaluation and the trace $H^2(W, \omega_W) \to \mathbb{C}$. Precomposition on the Hom term shows the required contravariance in $\mathcal{G}$. □

Corollary 6.8 (the duality bridge used in Section 10)**.** *Under the hypotheses of the proposition,*

$$H^0(W, \Omega^1_X|_W \otimes A_L) \neq 0 \quad \Longrightarrow \quad H^2(W, i^*\mathcal{F}_L) \neq 0.$$

Corollary 6.9 (candidate fibrewise nonvanishing)**.** ***[VERIFIED]*** *For every $L \in \mathrm{Pic}^0(X)$,*

$$H^2(W, i^*\mathcal{F}_L) \neq 0.$$

The duality isomorphism itself holds for every $L \in \mathrm{Pic}(X)$. If $r = 0$, the displayed nonvanishing also holds for every such L.

Proof. Proposition 6.3 verifies that W is the compact full Cartier fibre in the smooth threefold, so the abstract duality theorem applies. Proposition 6.5 supplies its nonzero dual section for $L \in \mathrm{Pic}^0(X)$. The final assertion uses Corollary 6.6. □

6.4 The exact top-degree base-change implication required

Proposition 6.10 (top-degree analytic base change)**.** ***[VERIFIED]*** *Let $f : X \to B$ be a proper morphism of complex spaces, $\mathcal{F}$ a coherent sheaf flat over B near $c \in B$, and X_c the analytic, hence full scheme-theoretic, fibre. If $H^3(X_c, \mathcal{F}|_{X_c}) = 0$, then the canonical map*

$$\beta^2_c : (R^2 f_*\mathcal{F})_c \otimes_{\mathcal{O}_{B,c}} \mathbb{C}(c) \xrightarrow{\sim} H^2(X_c, \mathcal{F}|_{X_c})$$

is an isomorphism. In particular, a nonzero target forces $(R^2 f_*\mathcal{F})_c \neq 0$.

Proof. Bănică–Stănăşilă's base-change theorem identifies surjectivity of the degree-q fibre map with right exactness of the corresponding direct-image functor. Their top-degree corollary states that if $H^q(X_c, \mathcal{F}|_{X_c}) = 0$, then these equivalent conditions hold in degree $q-1$, and the degree-$(q-1)$ base-change maps are isomorphisms [6, Chapter III, §3, Theorem 3.4 and Corollary 3.5, pp. 116–121]. Taking $q = 3$ gives the displayed map. If its target is nonzero, its source is nonzero; hence the stalk itself cannot vanish. □

Lemma 6.11 (automatic properness and flatness over the curve)**.** ***[VERIFIED]*** *Let X be a compact connected smooth complex manifold, let B be a smooth complex curve, and let $f : X \to B$ be nonconstant and holomorphic. Then f is proper and flat. Every locally free $\mathcal{O}_X$-module is f-flat, and every full scheme-theoretic fibre is an effective Cartier divisor of pure codimension one.*

Proof. The underlying continuous map is proper because X is compact and B is Hausdorff. Fix $x \in X$, put $b = f(x)$, and write $A = \mathcal{O}_{B,b}$ and $R = \mathcal{O}_{X,x}$. The ring A is a discrete valuation ring and R is a regular local analytic ring, hence an integral domain. The homomorphism $A \to R$ is injective: if a nonzero germ on the curve pulled back to zero near x, then a neighbourhood of x would map into its discrete zero set, so f would be locally constant and therefore, by connectedness and analytic continuation, globally constant.

Thus R is torsion-free as an A-module. A torsion-free module over a discrete valuation ring is flat, so f is flat at x; since x was arbitrary, f is flat. A locally free sheaf has stalk R^r, which is flat over A by the same argument. Finally, if t is a local parameter at b, its pullback is a nonzero element of the domain R and hence a non-zero-divisor. Therefore the full fibre is locally cut out by one non-zero-divisor: it is an effective Cartier divisor and is pure of codimension one. □

For a full fibre of complex dimension two and a coherent sheaf, $H^3 = 0$. Proposition 6.3 supplies the candidate geometry, and Lemma 6.11 makes the locally free sheaf $\mathcal{F}_L = T_X \otimes L$ flat over the base.

Corollary 6.12 (candidate base change and direct-image nonvanishing)**.** ***[VERIFIED]*** *For every* $L \in \mathrm{Pic}(X)$*, the canonical map*

$$(R^2 f_* \mathcal{F}_L)_{p_0} \otimes \mathbb{C}(p_0) \xrightarrow{\sim} H^2(W, i^* \mathcal{F}_L)$$

is an isomorphism. Consequently, for every $L \in \mathrm{Pic}^0(X)$*,*

$$(R^2 f_* \mathcal{F}_L)_{p_0} \neq 0, \qquad \textit{and hence} \qquad R^2 f_*(T_X \otimes L) \neq 0.$$

If $r = 0$*, the same nonvanishing holds for every* $L \in \mathrm{Pic}(X)$*.*

Proof. The candidate is proper and $\mathcal{F}_L$ is base-flat, while the two-dimensional fibre has $H^3(W, i^* \mathcal{F}_L) = 0$. Apply Proposition 6.10. The nonzero target for $L \in \mathrm{Pic}^0(X)$ is Corollary 6.9; the all-line-bundle extension is the conditional final clause of that corollary. □

6.5 Present evidentiary status

- The reconstructed Sections 3–6 establish the compact smooth analytic threefold, the proper map, the full reduced normal-crossings fibre, its dP_6 normalization, and the local triviality of ω_X near W. These claims have now been retraced independently from the frozen source; they use none of the open integral-topology maps.
- For every $L \in \mathrm{Pic}^0(X)$, the pullback of A_L to the normalization is trivial. The conductor section, compact-Cartier duality, and top-degree base change therefore prove $R^2 f_*(T_X \otimes L) \neq 0$ for every such L.
- Extending this conclusion to every $L \in \mathrm{Pic}(X)$ is a separate question: for this method it is equivalent to $r = 0$. The claimed $H^2(X; \mathbb{Z}) = 0$ would suffice, but still depends on the open topology bridges.
- The conductor section invalidates the printed cotangent-sheaf reduction as a formal inference on this non-normal fibre. This does not by itself decide whether CDP20, Theorem 2.2 has another proof or whether the candidate satisfies all of that theorem's global hypotheses.

6.6 Remaining verification and falsification programme

1. **All-line-bundle restriction map.** Compute $r = \eta^* i^*$ from explicit transition functions. A line bundle with $r(L) \neq 0$ refutes the manuscript's universal conductor argument but does not affect the proved Pic^0 result. A direct proof that $r = 0$ would extend the nonvanishing to all line bundles without borrowing the unresolved topology.
2. **Global CDP hypotheses.** Establish independently the exact topological vanishings, $a(X) = 1$, and the holomorphic-algebraic-reduction claim used to place this X under CDP20, Theorem 2.2. The analytic nonvanishing above is not a substitute for those global premises.
3. **Literature comparison.** Reproduce the CDP p. 692 argument with exactly the same fibre, twist, and definition of $\widetilde{\Omega}^1_S$. Determine whether an omitted hypothesis restores the inference or whether an independent proof bypasses it. Either outcome must be recorded without converting a proof-gap diagnosis into a theorem verdict.

Proof obligation 6.13 (CDP comparison)**.** **[OPEN]** Produce a line-by-line diagram comparing the candidate's ambient, conormal, and intrinsic cotangent sequences with the exact sheaves in the cited CDP reduction. Record separately: the validity of the printed inference, the possibility of an independent repair, and the validity of the candidate construction.

Disproof test 6.14 (decisive outcomes)**.** The proved Pic^0 chain can be refuted only by locating an error in one of its explicit inputs: the source construction of the full fibre, the normalization–conductor descent, the dP_6 Picard injection, compact-Cartier duality, or top-degree base change. The stronger all-line-bundle chain fails exactly when the restriction map r is nonzero. Completing proof obligation 6.13 would determine whether the printed CDP reduction admits a repair, but would still leave the global topology obligations C1–C3, V1, MV1, and L1.

7 Confirmed local repairs kept separate from the theorem

These corrections are mathematically definite but do not settle the central construction.

1. **[REFUTED]** Proposition 4.7(iv): the printed closedness justification is not valid as written. A repair would require a complete chartwise/local-finiteness argument, which remains **[OPEN]**.

2. **[REFUTED]** Remark 7.28 is false for a fixed triple. For example,

$$(\ell_0, \ell_1, \ell_2) = (100, 1, -1), \qquad p = 12(100) - 4(1) - 3(-1) = 1199 \neq \pm 1.$$

   The intended statement must quantify existentially over ℓ_0 for fixed admissible (ℓ_1, ℓ_2).

3. **[VERIFIED]** Appendix Lemma A.3 prints cohomological arrows while labelling them as homology. The sequence must be written

$$H^k(W) \longrightarrow H^k(\widetilde{W}) \oplus H^k(D) \longrightarrow H^k(\widetilde{D}) \longrightarrow H^{k+1}(W),$$

   followed by universal coefficients when homology ranks are required.

4. **[VERIFIED]** Appendix Remark A.6 must read $\pi_1(N_0) = \bar{\Lambda} = \pi_1(W)$, not $\pi_1(N_0) = \Lambda$.

5. **[VERIFIED]** The modular assembler has a page-71 portability defect under the tested MiKTEX route; the canonical monolithic source renders correctly. This is a build defect, not a mathematical counterexample.

8 Research log

Round 0 — private workbench initialization

- Preserved the completed 121-page annotated audit as `audit_baseline/annotated_audit.tex` and the 108-page reconstruction as `audit_baseline/clean_transcription.tex`.
- Converted the review objections into named mathematical obligations with explicit maps, bases, signs, and falsification tests.
- Proved the exact conditional endpoint: once the geometric maps are identified with the printed matrices, the verified algebra gives the standard smooth 6-sphere.
- Kept the conductor mechanism separate from both the existence of X and the truth or falsity of the published CDP theorem.

Next calculations

1. Extract the actual coordinate formulas for the three collar inclusions and build a complete orientation/sign table.

2. Build a finite integral chain model for the cusp central fibre W and compare its specialization map with Appendix B in primitive bases.

3. Compute the Leray d_2 geometrically and compare it to $d_2(12\gamma) = \pm p\omega$ without importing the Mayer–Vietoris normalization.

4. Compute the restriction map $r = \eta^* i^* : \mathrm{Pic}(X) \to \mathrm{Pic}(\mathrm{dP}_6)$ from transition functions, and complete the line-by-line CDP cotangent-sheaf comparison without using the unresolved topology.

Round 1 — 26 August 2026: typed workbench release

- Rebuilt the workbench as a new clean tree with the exact title "S6 Proof Repair and Disproof Workbench — Private". The public audit project is not an input to the upload workflow and remains untouched.

- Kept the terminal algebra **[VERIFIED]**, but kept the global topology **[CONDITIONAL]** until C1–C3, V1, MV1, and L1 identify the printed matrices with actual geometric maps. No theorem-level CDP verdict is inferred from a proof-step defect alone.

- Added a reader-facing literature audit with DOI-resolvable references for CDP20/CDP98, analytic duality and base change, nearby cycles, topology, and the Leiden Declaration. Local SGA TeX was used as a comparison layer; third-party files are not bundled.

- Added the bounded Fable/Jacobian provenance question to the comparative workstream. Any source path not independently located remains **[OPEN]** rather than being presented as a recovered transcript.

Round 2 — 26 August 2026: duality and base-change bridges

- Proved the abstract compact-Cartier-fibre duality statement from Ramis–Ruget's analytic dualizing complex and trace pairing, Cartier adjunction, and coherent-cohomology finiteness. Normality of the fibre is not required.

- Promoted the already complete normalization–conductor calculation to **[VERIFIED]** and separated it from its **[CONDITIONAL]** application to the candidate. Also isolated the elementary dP_6 Picard pullback lemma: $H^2(X;\mathbb{Z}) = 0$ forces every line bundle from X to pull back trivially to the normalization.

- At this round, promoted only the abstract theorem to **[VERIFIED]** and left its candidate application **[CONDITIONAL]** on smoothness, compactness, the full scheme-theoretic fibre, and Cartierness. Round 3 below supersedes that interim status after independently retracing Sections 3–6.

- Proved the required abstract top-degree base-change isomorphism from the degree-3 vanishing corollary of the analytic base-change theorem. Also proved that properness, flatness, Cartierness of the full fibre, and f-flatness of a vector bundle are automatic for a nonconstant map from a compact connected smooth complex manifold to a smooth curve.

- At this round, kept the candidate instantiation **[OPEN]** pending a chart audit. Round 3 below records the completed source reconstruction and status promotion.

- Reconciled the conditional topology endpoint with the six exact bridge obligations C1–C3, V1, MV1, and L1; no four-item shorthand now substitutes for that dependency set.

Round 3 — 26 August 2026: candidate geometry and the Pic^0 strengthening

- Independently retraced Theorem 3.4, Theorem 4.5, Proposition 4.6, Proposition 4.7(i)–(iii), Theorem 5.4, and Theorem 6.2 in the frozen transcription. This closes the analytic action, quotient, global gluing, full-fibre, dP_6 normalization, and local canonical-triviality gates. It does not close the cusp-collapse or integral topology maps.

- Proved a topology-free Picard lemma: for every $L \in \mathrm{Pic}^0(X)$, the line bundle $A_L = (L^\vee \otimes \omega_X)|_W$ pulls back trivially to dP_6. The proof uses $c_1(L) = 0$, injectivity of the Chern-class map on dP_6, and the nowhere-vanishing three-form on N_0.

- Combined that lemma with the verified conductor section, analytic duality, and top-degree base change to obtain $R^2 f_*(T_X \otimes L) \neq 0$ for every $L \in \mathrm{Pic}^0(X)$. This conclusion no longer borrows $H^2(X; \mathbb{Z}) = 0$.

- Isolated the exact remaining all-line-bundle question as $r = \eta^* i^* = 0$. The open global topology would imply it, but the Pic^0 conclusion needed for generic CDP twists is already independent of that route.

9 Literature audit and source gate

This is a publication-facing literature record, not a list of private chat notes. The CDP comparison is against Campana–Demailly–Peternell's 2020 paper [2, Theorem 2.2 and Corollary 2.3, p. 680], with the 1998 paper [3, Lemma 3.2, pp. 84–85] retained because the later argument imports its semidirect-product calculation. The exact theorem and page locators are given in the bibliography below; the DOI links are the reader-facing source gate.

For the analytic part, the workbench uses the absolute duality results of Ramis–Ruget [4, Theorems 1–2 and Proposition 2, pp. 81–91] and the relative analytic reference of Ramis–Ruget–Verdier [5], together with Bănică–Stănășilă's cohomology-and-base- change criteria [6]. The latter is cited at Chapter III, §3, Theorems 3.4 and Corollaries 3.5 and 3.10, and §4, Theorem 4.12. The Stacks Project is used only as an algebraic cross-check, not as a substitute for the analytic theorem. Nearby-cycle formalism is compared with SGA 7 I/II and with the analytic treatment cited in the bibliography; none of those general references identifies the candidate's printed matrices without a candidate-specific map calculation.

The local literature corpus was also checked before writing this section. The relevant SGA 6 and SGA 7 TeX paths are recorded verbatim in `research/README.md` and `comparative_context/README.md`; they were read as a local comparison layer. No third-party source files or PDFs are copied into this project. The public citations remain the canonical way for a reader to check the literature.

The mathematical role of each source is deliberately narrow:

- CDP20 supplies the theorem-level target and the published proof steps under audit; CDP98 supplies the earlier split-extension calculation.

- Ramis–Ruget, Ramis–Ruget–Verdier, and Bănică–Stănășilă supply the analytic duality and base-change statements whose hypotheses are still checked explicitly in Section 10.

- SGA 7 and the analytic nearby-cycle references supply formal frameworks [21, 22], not the candidate-specific integral identifications.

- Hatcher, Smale, and Kervaire–Milnor supply the final recognition theorems [9, 11, 12] only after the geometric maps have been certified.
- The Leiden Declaration is recorded as a methodological disclosure reference [1]; it is not mathematical evidence for any claim.

No literature result is silently promoted from a general theorem to a fact about the candidate. Any such promotion must appear as a typed claim in the ledgers and carry its own map, hypothesis, and falsification test.

References

[1] J. Alper, M. Barany, A. Chavarri Villarello, S. Dahmen, W. Dean, K. Ganapathy, M. Harris, D. Holmes, M. Jamnik, S. Kelk, B. Kra, U. Martin, B. Naskręcki, R. Ochigame, J. Portegies, and J. Schmitt, *Leiden Declaration on Artificial Intelligence and Mathematics*, 2026, doi:10.5281/zenodo.20302944.

[2] F. Campana, J.-P. Demailly, and T. Peternell, *The algebraic dimension of compact complex threefolds with vanishing second Betti number*, Compos. Math. **156** (2020), 679–696, doi:10.1112/S0010437X19007802.

[3] F. Campana, J.-P. Demailly, and T. Peternell, *The algebraic dimension of compact complex threefolds with vanishing second Betti number*, Compos. Math. **112** (1998), 77–91, doi:10.1023/A:1000313214795; arXiv:math/9607215.

[4] J.-P. Ramis and G. Ruget, *Complexe dualisant et théorèmes de dualité en géométrie analytique complexe*, Publ. Math. IHÉS **38** (1970), 77–91, doi:10.1007/BF02684652.

[5] J.-P. Ramis, G. Ruget, and J.-L. Verdier, *Dualité relative en géométrie analytique complexe*, Invent. Math. **13** (1971), 261–283, doi:10.1007/BF01406078; EuDML record.

[6] C. Bănică and O. Stănăşilă, *Algebraic Methods in the Global Theory of Complex Spaces*, Editura Academiei and John Wiley & Sons, Bucharest and London, 1976; Chapter III, §3, Theorem 3.4 and Corollaries 3.5 and 3.10, pp. 116–123; Chapter III, §4, Theorem 4.12, pp. 134–135, publicly accessible scan.

[7] H. Grauert, *Ein Theorem der analytischen Garbentheorie und die Modulräume komplexer Strukturen*, Publ. Math. IHÉS **5** (1960), 5–64, doi:10.1007/BF02684746.

[8] J. M. Lee, *Introduction to Complex Manifolds*, Graduate Studies in Mathematics **244**, American Mathematical Society, Providence, RI, 2024; Theorem 9.37, publisher page.

[9] A. Hatcher, *Algebraic Topology*, Cambridge University Press, Cambridge, 2002; Theorems 4.5 and 4.32 and Corollary 4.33, author's electronic edition.

[10] G. E. Bredon, *Sheaf Theory*, second edition, Graduate Texts in Mathematics **170**, Springer, New York, 1997; Chapter VI, doi:10.1007/978-1-4612-0647-7.

[11] S. Smale, *Generalized Poincaré's conjecture in dimensions greater than four*, Ann. of Math. (2) **74** (1961), 391–406, doi:10.2307/1970239.

[12] M. A. Kervaire and J. W. Milnor, *Groups of homotopy spheres: I*, Ann. of Math. (2) **77** (1963), 504–537, doi:10.2307/1970128.

[13] R. Hartshorne, *Algebraic Geometry*, Graduate Texts in Mathematics **52**, Springer, New York, 1977.

[14] C. Birkenhake and H. Lange, *Complex Tori*, Progress in Mathematics **177**, Birkhäuser, Boston, 1999; Chapter 2, §6, doi:10.1007/978-1-4612-1566-0.

[15] G. Pareschi, M. Popa, and C. Schnell, *Hodge modules on complex tori and generic vanishing for compact Kähler manifolds*, Geom. Topol. **21** (2017), 2419–2460; §8, p. 2437, doi:10.2140/gt.2017.21.2419.

[16] W. Fulton, *Introduction to Toric Varieties*, Annals of Mathematics Studies **131**, Princeton University Press, Princeton, 1993.

[17] D. A. Cox, J. B. Little, and H. K. Schenck, *Toric Varieties*, Graduate Studies in Mathematics **124**, American Mathematical Society, Providence, RI, 2011; Theorem 12.1.10, doi:10.1090/gsm/124.

[18] V. I. Danilov, *The geometry of toric varieties*, Russian Math. Surveys **33** (1978), no. 2, 97–154; §9, doi:10.1070/RM1978v033n02ABEH002305.

[19] J.-P. Serre, *A Course in Arithmetic*, Graduate Texts in Mathematics **7**, Springer, New York, 1973; Chapter VII, doi:10.1007/978-1-4684-9884-4.

[20] K. S. Brown, *Cohomology of Groups*, Graduate Texts in Mathematics **87**, Springer, New York, 1982.

[21] A. Grothendieck, M. Raynaud, and D. S. Rim (eds.), *Groupes de monodromie en géométrie algébrique. I*, Séminaire de Géométrie Algébrique du Bois-Marie 1967–1969, Lecture Notes in Mathematics **288**, Springer, Berlin, 1972, doi:10.1007/BFb0068688.

[22] P. Deligne and N. Katz (eds.), *Groupes de monodromie en géométrie algébrique. II*, Séminaire de Géométrie Algébrique du Bois-Marie 1967–1969, Lecture Notes in Mathematics **340**, Springer, Berlin, 1973.

[23] A. Dimca, *Sheaves in Topology*, Universitext, Springer, Berlin, 2004; Definition 4.2.1, Propositions 4.2.2 and 4.2.11, Theorem 2.3.26, and Corollary 2.3.4, doi:10.1007/978-3-642-18868-8.

[24] M. Kashiwara and P. Schapira, *Sheaves on Manifolds*, Grundlehren der mathematischen Wissenschaften **292**, Springer, Berlin, 1990; §8.6, doi:10.1007/978-3-662-02661-8.

[25] A. Grothendieck and the SGA contributors, *SGA in English: maintained working edition*, 2026, doi:10.5281/zenodo.21925497; source repository. This working edition is cited only as a searchable reading layer; the original volumes remain the authority.

[26] The Stacks Project Authors, *The Stacks Project*, stacks.math.columbia.edu, especially Tags 0AUW and 0EX2.